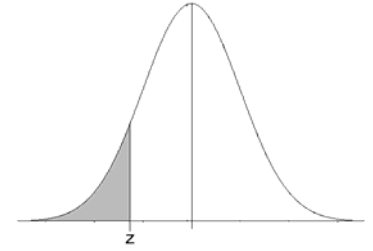


Standard Normal Cumulative Probability Table



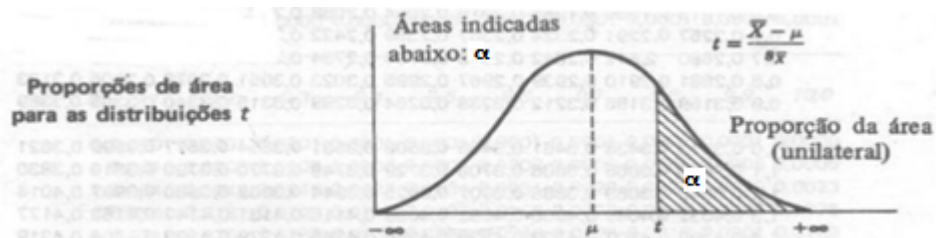
Cumulative probabilities for NEGATIVE z-values are shown in the following table:

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

A normal distribution curve is shown. The area under the curve to the left of a point labeled z on the horizontal axis is shaded gray. The point z is located to the right of the mean (the peak of the curve).

A normal distribution curve is shown. The area under the curve to the left of a point labeled z on the horizontal axis is shaded gray. The point z is located to the right of the mean (the peak of the curve).

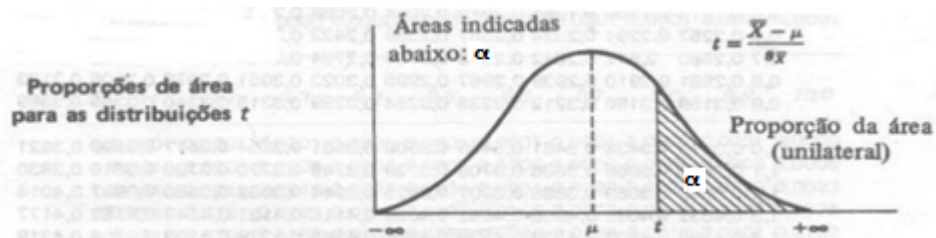
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998



	Nível de significância (α)				
G.L.	0,100	0,050	0,025	0,010	0,005
1	3,078	6,314	12,706	31,821	63,657
2	1,886	2,920	4,303	6,965	9,925
3	1,638	2,353	3,182	4,541	5,841
4	1,533	2,132	2,776	3,747	4,604
5	1,476	2,015	2,571	3,365	4,032
6	1,440	1,943	2,447	3,143	3,707
7	1,415	1,895	2,365	2,998	3,499
8	1,397	1,860	2,306	2,896	3,355
9	1,383	1,833	2,262	2,821	3,250
10	1,372	1,812	2,228	2,764	3,169
11	1,363	1,796	2,201	2,718	3,106
12	1,356	1,782	2,179	2,681	3,055
13	1,350	1,771	2,160	2,650	3,012
14	1,345	1,761	2,145	2,624	2,977
15	1,341	1,753	2,131	2,602	2,947
16	1,337	1,746	2,120	2,583	2,921
17	1,333	1,740	2,110	2,567	2,898
18	1,330	1,734	2,101	2,552	2,878
19	1,328	1,729	2,093	2,539	2,861
20	1,325	1,725	2,086	2,528	2,845
21	1,323	1,721	2,080	2,518	2,831
22	1,321	1,717	2,074	2,508	2,819
23	1,319	1,714	2,069	2,500	2,807
24	1,318	1,711	2,064	2,492	2,797
25	1,316	1,708	2,060	2,485	2,787
26	1,315	1,706	2,056	2,479	2,779
27	1,314	1,703	2,052	2,473	2,771
28	1,313	1,701	2,048	2,467	2,763
29	1,311	1,699	2,045	2,462	2,756
30	1,310	1,697	2,042	2,457	2,750
31	1,309	1,696	2,040	2,453	2,744
32	1,309	1,694	2,037	2,449	2,738
33	1,308	1,692	2,035	2,445	2,733
34	1,307	1,691	2,032	2,441	2,728
35	1,306	1,690	2,030	2,438	2,724
36	1,306	1,688	2,028	2,434	2,719
37	1,305	1,687	2,026	2,431	2,715
38	1,304	1,686	2,024	2,429	2,712

	Nível de significância (α)				
G.L.	0,1	0,05	0,025	0,01	0,005
39	1,304	1,685	2,023	2,426	2,708
40	1,303	1,684	2,021	2,423	2,704
41	1,303	1,683	2,020	2,421	2,701
42	1,302	1,682	2,018	2,418	2,698
43	1,302	1,681	2,017	2,416	2,695
44	1,301	1,680	2,015	2,414	2,692
45	1,301	1,679	2,014	2,412	2,690
46	1,300	1,679	2,013	2,410	2,687
47	1,300	1,678	2,012	2,408	2,685
48	1,299	1,677	2,011	2,407	2,682
49	1,299	1,677	2,010	2,405	2,680
50	1,299	1,676	2,009	2,403	2,678
51	1,298	1,675	2,008	2,402	2,676
52	1,298	1,675	2,007	2,400	2,674
53	1,298	1,674	2,006	2,399	2,672
54	1,297	1,674	2,005	2,397	2,670
55	1,297	1,673	2,004	2,396	2,668
56	1,297	1,673	2,003	2,395	2,667
57	1,297	1,672	2,002	2,394	2,665
58	1,296	1,672	2,002	2,392	2,663
59	1,296	1,671	2,001	2,391	2,662
60	1,296	1,671	2,000	2,390	2,660
61	1,296	1,670	2,000	2,389	2,659
62	1,295	1,670	1,999	2,388	2,657
63	1,295	1,669	1,998	2,387	2,656
64	1,295	1,669	1,998	2,386	2,655
65	1,295	1,669	1,997	2,385	2,654
66	1,295	1,668	1,997	2,384	2,652
67	1,294	1,668	1,996	2,383	2,651
68	1,294	1,668	1,995	2,382	2,650
69	1,294	1,667	1,995	2,382	2,649
70	1,294	1,667	1,994	2,381	2,648
71	1,294	1,667	1,994	2,380	2,647
72	1,293	1,666	1,993	2,379	2,646
73	1,293	1,666	1,993	2,379	2,645
74	1,293	1,666	1,993	2,378	2,644
75	1,293	1,665	1,992	2,377	2,643
76	1,293	1,665	1,992	2,376	2,642

	Nível de significância (α)				
G.L.	0,1	0,05	0,025	0,01	0,005
77	1,293	1,665	1,991	2,376	2,641
78	1,292	1,665	1,991	2,375	2,640
79	1,292	1,664	1,990	2,374	2,640
80	1,292	1,664	1,990	2,374	2,639
81	1,292	1,664	1,990	2,373	2,638
82	1,292	1,664	1,989	2,373	2,637
83	1,292	1,663	1,989	2,372	2,636
84	1,292	1,663	1,989	2,372	2,636
85	1,292	1,663	1,988	2,371	2,635
86	1,291	1,663	1,988	2,370	2,634
87	1,291	1,663	1,988	2,370	2,634
88	1,291	1,662	1,987	2,369	2,633
89	1,291	1,662	1,987	2,369	2,632
90	1,291	1,662	1,987	2,368	2,632
91	1,291	1,662	1,986	2,368	2,631
92	1,291	1,662	1,986	2,368	2,630
93	1,291	1,661	1,986	2,367	2,630
94	1,291	1,661	1,986	2,367	2,629
95	1,291	1,661	1,985	2,366	2,629
96	1,290	1,661	1,985	2,366	2,628
97	1,290	1,661	1,985	2,365	2,627
98	1,290	1,661	1,984	2,365	2,627
99	1,290	1,660	1,984	2,365	2,626
100	1,290	1,660	1,984	2,364	2,626
101	1,290	1,660	1,984	2,364	2,625
102	1,290	1,660	1,983	2,363	2,625
103	1,290	1,660	1,983	2,363	2,624
104	1,290	1,660	1,983	2,363	2,624
105	1,290	1,659	1,983	2,362	2,623
106	1,290	1,659	1,983	2,362	2,623
107	1,290	1,659	1,982	2,362	2,623
108	1,289	1,659	1,982	2,361	2,622
109	1,289	1,659	1,982	2,361	2,622
110	1,289	1,659	1,982	2,361	2,621
111	1,289	1,659	1,982	2,360	2,621
112	1,289	1,659	1,981	2,360	2,620
113	1,289	1,658	1,981	2,360	2,620
114	1,289	1,658	1,981	2,360	2,620



	Nível de significância (α)				
G.L.	0,100	0,050	0,025	0,010	0,005
115	1,289	1,658	1,981	2,359	2,619
116	1,289	1,658	1,981	2,359	2,619
117	1,289	1,658	1,980	2,359	2,619
118	1,289	1,658	1,980	2,358	2,618
119	1,289	1,658	1,980	2,358	2,618
120	1,289	1,658	1,980	2,358	2,617
121	1,289	1,658	1,980	2,358	2,617
122	1,289	1,657	1,980	2,357	2,617
123	1,288	1,657	1,979	2,357	2,616
124	1,288	1,657	1,979	2,357	2,616
125	1,288	1,657	1,979	2,357	2,616
126	1,288	1,657	1,979	2,356	2,615
127	1,288	1,657	1,979	2,356	2,615
128	1,288	1,657	1,979	2,356	2,615
129	1,288	1,657	1,979	2,356	2,614
130	1,288	1,657	1,978	2,355	2,614
131	1,288	1,657	1,978	2,355	2,614
132	1,288	1,656	1,978	2,355	2,614
133	1,288	1,656	1,978	2,355	2,613
134	1,288	1,656	1,978	2,354	2,613
135	1,288	1,656	1,978	2,354	2,613
136	1,288	1,656	1,978	2,354	2,612
137	1,288	1,656	1,977	2,354	2,612
138	1,288	1,656	1,977	2,354	2,612
139	1,288	1,656	1,977	2,353	2,612
140	1,288	1,656	1,977	2,353	2,611
141	1,288	1,656	1,977	2,353	2,611
142	1,288	1,656	1,977	2,353	2,611
143	1,287	1,656	1,977	2,353	2,611
144	1,287	1,656	1,977	2,353	2,610
145	1,287	1,655	1,976	2,352	2,610
146	1,287	1,655	1,976	2,352	2,610
147	1,287	1,655	1,976	2,352	2,610
148	1,287	1,655	1,976	2,352	2,609
149	1,287	1,655	1,976	2,352	2,609
150	1,287	1,655	1,976	2,351	2,609
151	1,287	1,655	1,976	2,351	2,609
152	1,287	1,655	1,976	2,351	2,609

	Nível de significância (α)				
G.L.	0,1	0,05	0,025	0,01	0,005
153	1,287	1,655	1,976	2,351	2,608
154	1,287	1,655	1,975	2,351	2,608
155	1,287	1,655	1,975	2,351	2,608
156	1,287	1,655	1,975	2,350	2,608
157	1,287	1,655	1,975	2,350	2,608
158	1,287	1,655	1,975	2,350	2,607
159	1,287	1,654	1,975	2,350	2,607
160	1,287	1,654	1,975	2,350	2,607
161	1,287	1,654	1,975	2,350	2,607
162	1,287	1,654	1,975	2,350	2,607
163	1,287	1,654	1,975	2,349	2,606
164	1,287	1,654	1,975	2,349	2,606
165	1,287	1,654	1,974	2,349	2,606
166	1,287	1,654	1,974	2,349	2,606
167	1,287	1,654	1,974	2,349	2,606
168	1,287	1,654	1,974	2,349	2,605
169	1,287	1,654	1,974	2,349	2,605
170	1,287	1,654	1,974	2,348	2,605
171	1,287	1,654	1,974	2,348	2,605
172	1,286	1,654	1,974	2,348	2,605
173	1,286	1,654	1,974	2,348	2,605
174	1,286	1,654	1,974	2,348	2,604
175	1,286	1,654	1,974	2,348	2,604
176	1,286	1,654	1,974	2,348	2,604
177	1,286	1,654	1,973	2,348	2,604
178	1,286	1,653	1,973	2,347	2,604
179	1,286	1,653	1,973	2,347	2,604
180	1,286	1,653	1,973	2,347	2,603
181	1,286	1,653	1,973	2,347	2,603
182	1,286	1,653	1,973	2,347	2,603
183	1,286	1,653	1,973	2,347	2,603
184	1,286	1,653	1,973	2,347	2,603
185	1,286	1,653	1,973	2,347	2,603
186	1,286	1,653	1,973	2,347	2,603
187	1,286	1,653	1,973	2,346	2,602
188	1,286	1,653	1,973	2,346	2,602
189	1,286	1,653	1,973	2,346	2,602
190	1,286	1,653	1,973	2,346	2,602

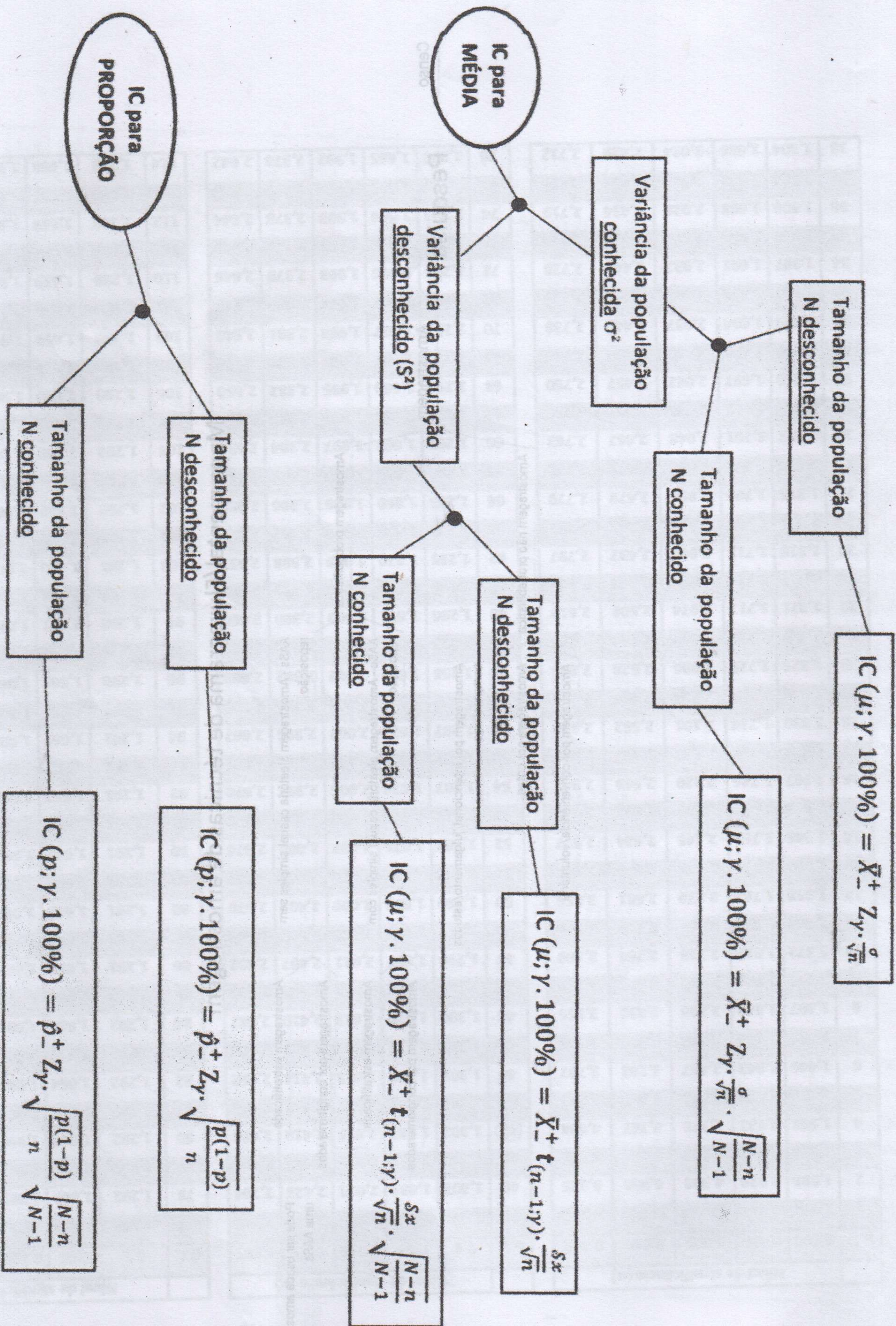
	Nível de significância (α)				
G.L.	0,1	0,05	0,025	0,01	0,005
191	1,286	1,653	1,972	2,346	2,602
192	1,286	1,653	1,972	2,346	2,602
193	1,286	1,653	1,972	2,346	2,602
194	1,286	1,653	1,972	2,346	2,601
195	1,286	1,653	1,972	2,346	2,601
196	1,286	1,653	1,972	2,346	2,601
197	1,286	1,653	1,972	2,345	2,601
198	1,286	1,653	1,972	2,345	2,601
199	1,286	1,653	1,972	2,345	2,601
200	1,286	1,653	1,972	2,345	2,601
201	1,286	1,652	1,972	2,345	2,601
202	1,286	1,652	1,972	2,345	2,600
203	1,286	1,652	1,972	2,345	2,600
204	1,286	1,652	1,972	2,345	2,600
205	1,286	1,652	1,972	2,345	2,600
206	1,286	1,652	1,972	2,345	2,600
207	1,286	1,652	1,971	2,344	2,600
208	1,286	1,652	1,971	2,344	2,600
209	1,286	1,652	1,971	2,344	2,600
210	1,286	1,652	1,971	2,344	2,599
211	1,286	1,652	1,971	2,344	2,599
212	1,286	1,652	1,971	2,344	2,599
213	1,286	1,652	1,971	2,344	2,599
214	1,286	1,652	1,971	2,344	2,599
215	1,286	1,652	1,971	2,344	2,599
216	1,285	1,652	1,971	2,344	2,599
217	1,285	1,652	1,971	2,344	2,599
218	1,285	1,652	1,971	2,344	2,599
219	1,285	1,652	1,971	2,343	2,598
220	1,285	1,652	1,971	2,343	2,598
221	1,285	1,652	1,971	2,343	2,598
222	1,285	1,652	1,971	2,343	2,598
223	1,285	1,652	1,971	2,343	2,598
224	1,285	1,652	1,971	2,343	2,598
225	1,285	1,652	1,971	2,343	2,598
500	1,283	1,648	1,965	2,334	2,586
1000	1,282	1,646	1,962	2,330	2,581
∞	1,282	1,645	1,960	2,326	2,576

$$Z \sim N(0, 1)$$

A graph of a normal distribution curve. The horizontal axis is labeled 'Z' and the vertical axis is labeled 'P'. The area under the curve between Z=0 and Z=Z_c is shaded and labeled 'P'.

parte inteira e primeira decimal de Z_c	Segunda decimal de Z_c										parte inteira e primeira decimal de Z_c
	0	1	2	3	4	5	6	7	8	9	
	p = 0										
0,0	00000	00399	00798	01197	01595	01994	02392	02790	03188	03586	0,0
0,1	03983	04380	04776	05172	05567	05962	06356	06749	07142	07535	0,1
0,2	07926	08317	08706	09095	09483	09871	10257	10642	11026	11409	0,2
0,3	11791	12172	12552	12930	13307	13683	14058	14431	14803	15173	0,3
0,4	15542	15910	16276	16640	17003	17364	17724	18082	18439	18793	0,4
0,5	19146	19497	19847	20194	20540	20884	21226	21566	21904	22240	0,5
0,6	22575	22907	23237	23565	23891	24215	24537	24857	25175	25490	0,6
0,7	25804	26115	26424	26730	27035	27337	27637	27935	28230	28524	0,7
0,8	28814	29103	29389	29673	29955	30234	30511	30785	31057	31327	0,8
0,9	31594	31859	32121	32381	32639	32894	33147	33398	33646	33891	0,9
1,0	34134	34375	34614	34850	35083	35314	35543	35769	35993	36214	1,0
1,1	36433	36650	36864	37076	37286	37493	37698	37900	38100	38298	1,1
1,2	38493	38686	38877	39065	39251	39435	39617	39796	39973	40147	1,2
1,3	40320	40490	40658	40824	40988	41149	41309	41466	41621	41774	1,3
1,4	41924	42073	42220	42364	42507	42647	42786	42922	43056	43189	1,4
1,5	43319	43448	43574	43699	43822	43943	44062	44179	44295	44408	1,5
1,6	44520	44630	44738	44845	44950	45053	45154	45254	45352	45449	1,6
1,7	45543	45637	45728	45818	45907	45994	46080	46164	46246	46327	1,7
1,8	46407	46485	46562	46638	46712	46784	46856	46926	46995	47062	1,8
1,9	47128	47193	47257	47320	47381	47441	47500	47558	47615	47670	1,9
2,0	47725	47778	47831	47882	47932	47982	48030	48077	48124	48169	2,0
2,1	48214	48257	48300	48341	48382	48422	48461	48500	48537	48574	2,1
2,2	48610	48645	48679	48713	48745	48778	48809	48840	48870	48899	2,2
2,3	48928	48956	48983	49010	49036	49061	49086	49111	49134	49158	2,3
2,4	49180	49202	49224	49245	49266	49286	49305	49324	49343	49361	2,4
2,5	49379	49396	49413	49430	49446	49461	49477	49492	49506	49520	2,5
2,6	49534	49547	49560	49573	49585	49598	49609	49621	49632	49643	2,6
2,7	49653	49664	49674	49683	49693	49702	49711	49720	49728	49736	2,7
2,8	49744	49752	49760	49767	49774	49781	49788	49795	49801	49807	2,8
2,9	49813	49819	49825	49831	49836	49841	49846	49851	49856	49861	2,9
3,0	49865	49869	49874	49878	49882	49886	49889	49893	49897	49900	3,0
3,1	49903	49906	49910	49913	49916	49918	49921	49924	49926	49929	3,1
3,2	49931	49934	49936	49938	49940	49942	49944	49946	49948	49950	3,2
3,3	49952	49953	49955	49957	49958	49960	49961	49962	49964	49965	3,3
3,4	49966	49968	49969	49970	49971	49972	49973	49974	49975	49976	3,4
3,5	49977	49978	49978	49979	49980	49981	49981	49982	49983	49983	3,5
3,6	49984	49985	49985	49986	49986	49987	49987	49988	49988	49989	3,6
3,7	49989	49990	49990	49990	49991	49991	49992	49992	49992	49992	3,7
3,8	49993	49993	49993	49994	49994	49994	49994	49995	49995	49995	3,8
3,9	49995	49995	49996	49996	49996	49996	49996	49996	49997	49997	3,9
4,0	49997	49997	49997	49997	49997	49997	49998	49998	49998	49998	4,0
4,5	49999	50000	50000	50000	50000	50000	50000	50000	50000	50000	4,5

Árvore de decisão para escolha do intervalo de confiança



Se $n > 30$ na maioria das vezes pode-se trocar $t_{(n-1; \gamma)}$ por Z_γ

Trocar a distribuição de referência t-de-student pela distribuição normal

Formulário:

1. Coeficiente de correlação linear de Pearson:

$$r = \frac{\sum_{i=1}^n x'_i y'_i}{n-1}$$

em que,

$$x'_i = \frac{x_i - \bar{x}}{S_x} \quad y'_i = \frac{y_i - \bar{y}}{S_y} \quad (i = 1, 2, \dots, n)$$

2. Coeficiente de correlação linear de Pearson (fórmula alternativa):

$$r = \frac{n \sum_{i=1}^n x_i y_i - (\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{\sqrt{n \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2} \sqrt{n \sum_{i=1}^n y_i^2 - (\sum_{i=1}^n y_i)^2}}$$

3. Inferência sobre ρ :

Supondo X e Y com distribuições normais, podemos realizar o teste, calculando

$$t = \frac{r \sqrt{n-2}}{\sqrt{1-r^2}}$$

e usando como distribuição de referência a t de Student com $gl = n - 2$.

4. Estimadores dos coeficientes linear (a) e angular (b):

$$\hat{b} = \frac{n \sum (x_i y_i) - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2}$$

$$\hat{a} = \frac{\sum y_i - \hat{b} \sum x_i}{n}$$

em que $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ é uma amostra observada.

E a chamada equação (reta) de regressão é dada por:

$$\hat{y} = \hat{a} + \hat{b}x$$

5. Teste t equivalente ao teste F , para testar se o coeficiente angular b é significativo:

$$t_c = \frac{\hat{b}}{s_{\hat{b}}}$$

Em que,

$$s_{\hat{e}} = \sqrt{\frac{(\sum y^2) - a(\sum y) - b(\sum xy)}{n-2}}$$

$$s_{\hat{b}} = s_{\hat{e}} \sqrt{\frac{1}{\sum x^2 - \left[\frac{(\sum x)^2}{n}\right]}}$$

e usando como distribuição de referência a t de Student com $gl = n - 2$.

6. Relação entre o teste t e F na análise de regressão.

$$t^2 = \frac{\hat{b}^2}{s_{\hat{b}}^2} = \frac{SQR}{\frac{SQE}{n-2}} = F$$

		População	
		Infinita	Finita
A. Estimativa de médias			
Pontual	$\bar{X}$	$\bar{X}$	$\bar{X}$
Intervalar			
σ_X conhecido	$\bar{X} \pm z_\gamma \frac{\sigma_X}{\sqrt{n}}$		$\bar{X} \pm z_\gamma \frac{\sigma_X}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$
σ_X desconhecido	$\bar{X} \pm t_{(\gamma, n-1)} \frac{S_X}{\sqrt{n}}$		$\bar{X} \pm t_{(\gamma, n-1)} \frac{S_X}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$
Tamanho da amostra			
σ_X conhecido	$n = \frac{z_\gamma^2 \sigma_X^2}{\epsilon^2}$		$n = \frac{z_\gamma^2 \sigma_X^2 N}{z_\gamma^2 \sigma_X^2 + \epsilon^2 (N-1)}$
σ_X desconhecido	$n = \frac{z_\gamma^2 S_X^2}{\epsilon^2}$		$n = \frac{z_\gamma^2 S_X^2 N}{z_\gamma^2 S_X^2 + \epsilon^2 (N-1)}$
Erro			
σ_X conhecido	$\epsilon = z_\gamma \frac{\sigma_X}{\sqrt{n}}$		$\epsilon = z_\gamma \frac{\sigma_X}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$
σ_X desconhecido	$\epsilon = t_{(\gamma, n-1)} \frac{S_X}{\sqrt{n}}$		$\epsilon = t_{(\gamma, n-1)} \frac{S_X}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$

		População	
		Infinita	Finita
B. Estimativa de proporções			
Pontual	$\hat{P}$	$\hat{P}$	$\hat{P}$
Intervalar			
Tamanho da amostra			
	$\hat{P} \pm z_\gamma \sqrt{\frac{\hat{P}(1-\hat{P})}{n}}$		$\hat{P} \pm z_\gamma \sqrt{\frac{\hat{P}(1-\hat{P})}{n}} \sqrt{\frac{N-n}{N-1}}$
	$n = \frac{z_\gamma^2 \hat{P}(1-\hat{P})}{\epsilon^2}$		$n = \frac{z_\gamma^2 \hat{P}(1-\hat{P}) N}{z_\gamma^2 \hat{P}(1-\hat{P}) + \epsilon^2 (N-1)}$
Erro			
	$\epsilon = z_\gamma \sqrt{\frac{\hat{P}(1-\hat{P})}{n}}$		$\epsilon = z_\gamma \sqrt{\frac{\hat{P}(1-\hat{P})}{n}} \sqrt{\frac{N-n}{N-1}}$

Mapa mental/Fluxograma de técnicas de amostragem

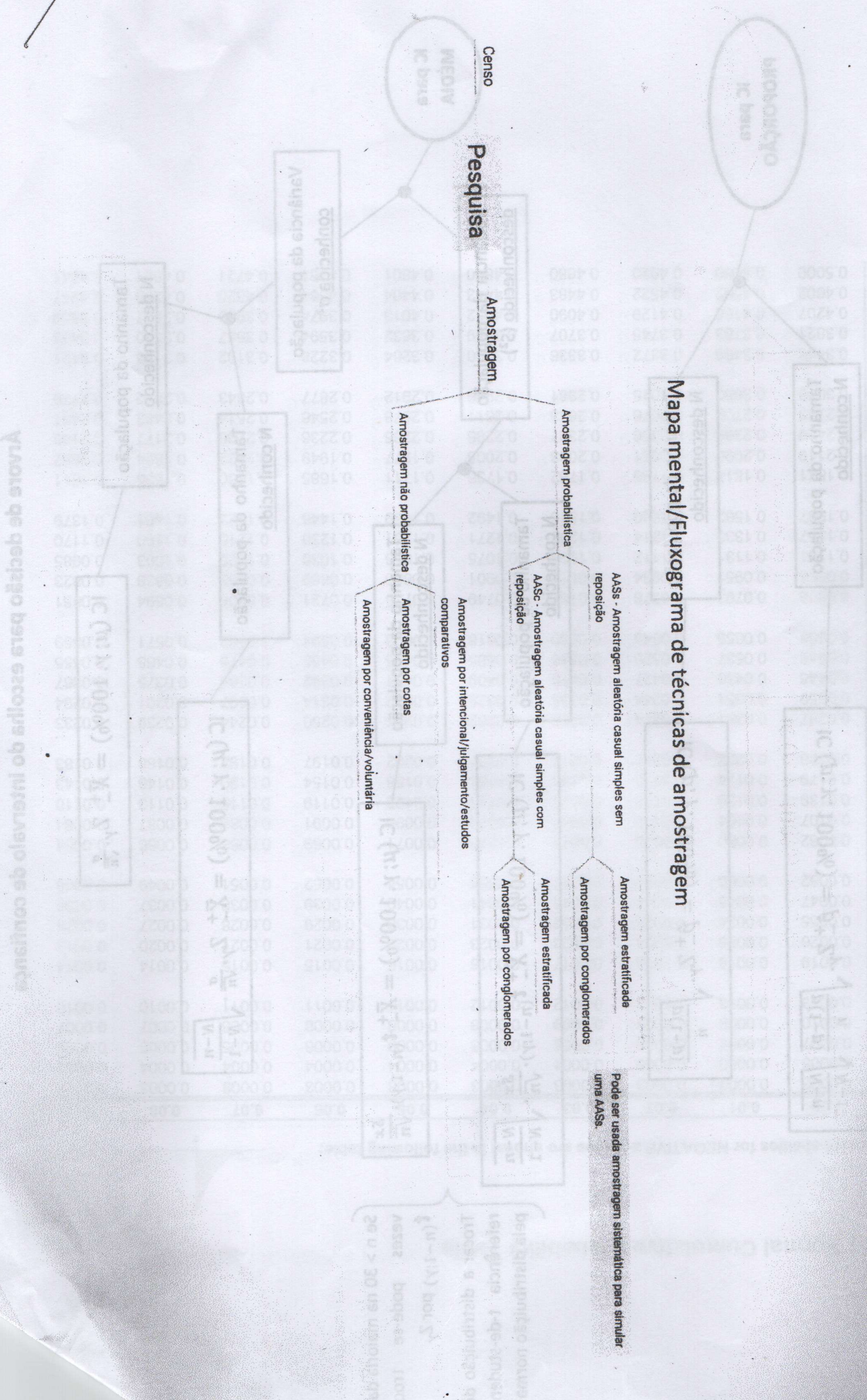
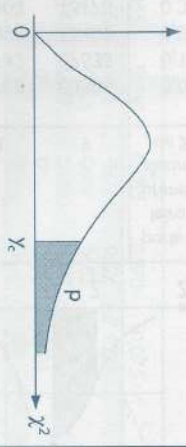


Tabela IV — Distribuição Qui-quadrado
 $Y \sim \chi^2 (v)$
Corpo da tabela dá os valores Y_c tais que $P(Y > Y_c) = p$.
Para valores $v > 30$, use a aproximação normal dada no texto.



p = 99%		98%	97,5%	95%	90%	80%	70%	50%	30%	20%	10%	5%	4%	2,5%	2%	1%	0,2%	0,1%	
1	0,016	0,063	0,001	0,004	0,016	0,064	0,148	0,455	1,074	1,642	2,706	3,841	4,218	5,024	5,412	6,635	9,550	10,827	1
2	0,020	0,040	0,051	0,103	0,211	0,446	0,713	1,386	2,408	3,219	4,605	5,991	6,438	7,378	7,824	9,210	12,429	13,815	2
3	0,115	0,185	0,216	0,352	0,584	1,005	1,424	2,366	3,665	4,642	6,251	7,815	8,311	9,348	9,837	11,345	14,796	16,266	3
4	0,297	0,429	0,484	0,711	1,064	1,649	2,195	3,357	4,878	5,989	7,779	9,488	10,026	11,143	11,668	13,277	16,924	18,467	4
5	0,554	0,752	0,831	1,145	1,610	2,343	3,000	4,351	6,064	7,289	9,236	11,070	11,644	12,832	13,388	15,086	18,907	20,515	5
6	0,872	1,134	1,237	1,635	2,204	3,070	3,828	5,348	7,231	8,558	10,645	12,592	13,198	14,449	15,033	16,812	20,791	22,457	6
7	1,239	1,564	1,690	2,167	2,833	3,822	4,671	6,346	8,383	9,803	12,017	14,067	14,703	16,013	16,622	18,475	22,601	24,322	7
8	1,646	2,032	2,180	2,733	3,490	4,594	5,527	7,344	9,524	10,656	12,242	14,684	15,307	16,517	17,171	19,023	23,090	24,352	8
9	2,088	2,532	2,700	3,325	4,168	5,380	6,393	8,343	10,656	12,242	14,684	16,919	17,608	19,023	20,483	21,161	23,209	26,056	9
10	2,558	3,059	3,247	3,940	4,865	6,179	7,267	9,342	11,781	13,442	15,987	18,307	19,021	20,483	21,161	23,209	27,722	29,588	10
11	3,053	3,609	3,816	4,575	5,578	6,989	8,148	10,341	12,899	14,631	17,275	19,675	20,412	21,920	22,618	24,725	29,354	31,264	11
12	3,571	4,178	4,404	5,226	6,304	7,807	9,034	11,340	14,011	15,812	18,549	21,026	21,785	23,337	24,054	26,217	30,957	32,909	12
13	4,107	4,765	5,009	5,892	7,042	8,634	9,926	12,340	15,119	16,985	19,812	22,362	23,142	24,736	25,472	27,688	32,535	34,528	13
14	4,660	5,368	5,629	6,571	7,790	9,467	10,821	13,339	16,222	18,151	21,064	23,685	24,485	26,119	26,873	29,141	34,091	36,123	14
15	5,229	5,985	6,262	7,261	8,547	10,307	11,721	14,339	17,322	19,311	22,307	24,996	25,816	27,488	28,259	30,578	35,628	37,697	15
16	5,812	6,614	6,908	7,962	9,312	11,152	12,624	15,338	18,418	20,465	23,542	26,296	27,136	28,845	29,633	32,000	37,146	39,252	16
17	6,408	7,255	7,564	8,672	10,085	12,002	13,531	16,338	19,511	21,615	24,769	27,587	28,445	30,191	30,995	33,409	38,468	40,790	17
18	7,015	7,906	8,231	9,390	10,865	12,857	14,440	17,338	20,601	22,760	25,989	28,869	29,745	31,526	32,346	34,805	40,136	42,312	18
19	7,633	8,567	8,906	10,117	11,651	13,716	15,352	18,338	21,689	23,900	27,204	30,144	31,037	32,852	33,687	36,191	41,610	43,820	19
20	8,260	9,237	9,591	10,851	12,443	14,578	16,266	19,337	22,775	25,038	28,412	31,410	32,321	34,170	35,020	37,566	43,072	45,315	20
21	8,897	9,915	10,283	11,591	13,240	15,445	17,182	20,337	23,858	26,171	29,615	32,671	33,597	35,479	36,343	38,932	44,522	46,797	21
22	9,542	10,600	10,982	12,338	14,041	16,314	18,101	21,337	24,939	27,301	30,813	33,924	34,867	36,781	37,659	40,289	45,962	48,268	22
23	10,196	11,293	11,688	13,091	14,848	17,187	19,021	22,337	26,018	28,429	32,007	35,172	36,131	38,076	38,968	41,638	47,391	49,728	23
24	10,856	11,992	12,401	13,848	15,659	18,062	19,943	23,337	27,096	29,553	33,196	36,415	37,389	39,364	40,270	42,980	48,812	51,179	24
25	11,524	12,697	13,120	14,611	16,473	18,940	20,867	24,337	28,172	30,655	34,382	37,652	38,642	40,646	41,566	44,314	50,223	52,620	25
26	12,198	13,409	13,844	15,379	17,292	19,820	21,792	25,336	29,246	31,795	35,563	38,885	39,889	41,923	42,856	45,642	51,627	54,052	26
27	12,879	14,125	14,573	16,151	18,114	20,703	22,719	26,336	30,319	32,912	36,741	40,113	41,132	43,194	44,140	46,963	53,022	55,476	27
28	13,565	14,847	15,308	16,928	18,939	21,588	23,647	27,336	31,319	33,927	37,916	41,337	42,370	44,461	45,419	48,278	54,411	56,893	28
29	14,258	15,574	16,047	17,708	19,768	22,475	24,577	28,336	32,461	35,139	39,087	42,557	43,604	45,722	46,693	49,588	55,792	58,302	29
30	14,953	16,306	16,791	18,493	20,599	23,364	25,508	29,336	33,530	36,250	40,266	43,773	44,834	46,979	47,962	50,892	57,167	59,703	30
p = 99%		98%	97,5%	95%	90%	80%	70%	50%	30%	20%	10%	5%	4%	2,5%	2%	1%	0,2%	0,1%	